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Halley6 Reserved Memory Configuration Description

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Release history

Date	Revision	Change
2022.12.22	2.0	First release

Disclaimer

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CATALOG

Key words	compile、burn、test
Abstract	
Abbreviations	
Reference material	《02-Halley6 Kernel development Manual_v2》

1 Reserved Memory Introduction

The reserved memory mainly solves the problem that some hardware devices need to repeatedly apply for a large amount of continuous physical memory.If system memory allocation is used, memory fragmentation may occur and memory application may fail.

By using the reserved memory allocation policy provided by Linux and estimating in advance, you can ensure that the hardware driver can obtain memory. The reserved memory is not managed by the system memory. After the reserved memory is configured in the DTS, the driver can apply for and release memory using standard apis.

2 Reserved Memory Configuration

2.1 Devicetree configuration description

The dts file mainly describes the start address and size of the reserved memory, and defines whether the device uses the reserved memory.

2.1.1 Default configuration

```
/ {
        reserved-memory {
                #address-cells = <1>;
                #size-cells = <1>;
                ranges =<>;

                reserved_memory: reserved_mem@0x2000000{
                      compatible = "shared-dma-pool";
                      reg = <0x02000000 0x2000000>;
                };
        };
};

&cim {
        status = "okay";
        memory-region=<&reserved_memory>;
};
```

2.1.2 Custom configuration

Users can modify it as required:

Start address, such as: 0x02000000

Size, such as: 0x2000000

The total memory size of the x1600 chip is 32M, and that of the x1600e chip is 64M. The reserved memory size should be appropriate and reasonable. Excessive memory size may cause kernel startup failure.

```
reserved_memory: reserved_mem@0x2000000{
                        compatible = "shared-dma-pool";
                        reg = <0x02000000 0x2000000>;
                };
```

2.2 Kernel driver configuration description

2.3 Reserved memory recommended configuration

According to the actual application scenario, the reserved memory mode is used for Camera and h264 codec applications, which can effectively reduce the problem of memory failure caused by memory fragmentation. Using the x1600 and x1600e as an example, the following configurations are recommended:

	DDR	reserved_mem	start addr	size
x1600	32M	8M	0x1800000	0x800000
x1600e	64M	8M	0x3800000	0x800000